

AHMED MOHAMED EBAID

Cloud Security / Security Engineer

☎ +20 155 339 5331 ✉ ahmedebaid0xc@gmail.com 📍 Giza, Egypt
🌐 linkedin.com/in/ahmed-ebaid-0xc 🐙 github.com/AhmadEbaid001

SUMMARY

Security-focused IT graduate specializing in cloud infrastructure security, with hands-on experience designing hybrid cloud architectures on AWS and Azure. Proficient in vulnerability assessment, SIEM operations, DevSecOps pipelines, and network security. Seeking a cloud security role securing cloud-native and hybrid environments.

EDUCATION

B.Sc. in Computational Science and Artificial Intelligence Oct 2022 – Jul 2026 (graduated)
Zewail City of Science and Technology Giza, Egypt

- Major: Information Technology. Concentration: Networks, Security and Governance.
- Relevant coursework: Network Security, Cloud Computing, IT Governance, Cryptography, Operating Systems, Database Systems.

PROFESSIONAL EXPERIENCE

Security Intern Jul 2025 – Aug 2025
Commercial International Bank (CIB) Egypt

- Conducted a vulnerability assessment simulation using **Nessus**, scanning a deliberately vulnerable endpoint, analyzing findings, and documenting remediation recommendations.
- Gained operational exposure to **Splunk SIEM**: explored log ingestion pipelines, constructed search queries, and configured a correlation alert for anomalous activity detection.
- Studied enterprise-grade security controls and governance frameworks applied in a regulated banking environment.

Network Intern Jul 2024 – Jan 2025
Zewail City IT Department Giza, Egypt

- Designed and partially implemented a network quota management system using open-source tools, enforcing per-user bandwidth policies across the campus network.
- Assisted in vulnerability assessments and security hardening tasks across the university internal network infrastructure.
- Participated in network installation and maintenance projects, gaining hands-on experience with enterprise switching, routing, and cabling.

PROJECTS

EcoForge GEMP: Green Energy Monitoring Platform Aug 2026 – Sep 2026
Python, FastAPI, Docker, TimescaleDB, MQTT, OR-Tools CP-SAT, nginx

- **Designed and implemented the full system** as sole engineer on a two-person team: architecture, telemetry ingestion, optimization engine, API, and front end, delivered as a containerized stack behind **457 automated tests**.
- Engineered **tamper-evident telemetry storage** with **HMAC-signed, hash-chained** meter readings and chain heads anchored outside the database volume, plus **Role-Based Access Control** and an audit log recording denied as well as permitted actions.
- Built a **DevSecOps CI/CD pipeline** blocking on any finding from **Bandit, Semgrep, pip-audit, Gitleaks, Hadolint, and Trivy** (filesystem, config, IaC, and image), with signed container builds and expiry-dated security exceptions.

AURA: Hybrid Cloud University Registration System Sep 2025 – Jun 2026
AWS, Azure, React.js, Express.js, PostgreSQL

- Architected a hybrid cloud solution integrating on-premises servers with **AWS** (IAM, VPC, Security Groups, Load Balancer) and **Azure** containerized microservices to handle registration traffic spikes.
- Designed and enforced **Role-Based Access Control (RBAC)** and OTP-based multi-factor authentication to protect student and faculty data.
- Configured **AWS IAM** policies with least-privilege principles and **Security Groups** to segment traffic between application tiers.
- Leveraged Azure container orchestration to enable auto-scaling during peak load, ensuring availability without over-provisioning.

Secure Communication Protocol May 2025
Python, PyCryptodome, RSA-2048, AES-GCM

- Built a secure messaging application implementing **RSA-2048** for key exchange, **AES-GCM** for authenticated encryption, and **RSA-PSS** digital signatures for non-repudiation.
- Engineered session key derivation using **HKDF-SHA256**, ensuring forward secrecy across communication sessions.

Automated Network Backup and Disaster Recovery Framework

Dec 2025

Bash, OpenSSL AES-256, Cron, EVE-NG

- Developed a zero-touch automated backup system achieving a **24-hour Recovery Point Objective (RPO)**, with AES-256 encryption at rest and SSH/RSA-2048 secured transfers.
- Validated network resilience via Link Aggregation (Bonding), achieving **148 Mbps throughput with zero packet loss**.

Enterprise Data Center Network Design

Dec 2025

Cisco Packet Tracer, OSPF, VLANs, SNMP

- Designed a three-tier hierarchical network supporting **600+ users** across five departments, with OSPF dynamic routing and Dual-ISP redundancy for high availability.
- Implemented VLSM for efficient IP allocation and SNMPv2c for centralized proactive monitoring.

TECHNICAL SKILLS

Cloud and Infrastructure: AWS (IAM, VPC, Security Groups, Load Balancer), Microsoft Azure (Containers, Auto-Scaling), Hybrid Cloud Architecture, Docker, Docker Compose, nginx

Security Tools: Nessus, Splunk (SIEM), Wireshark, Nmap, Snort, Volatility, Autopsy

DevSecOps: CI/CD security gating, GitHub Actions, SAST (Bandit, Semgrep), Software Composition Analysis (pip-audit), Secrets Scanning (Gitleaks), Container and IaC Scanning (Trivy, Hadolint), Container Image Signing

Security Domains: Vulnerability Assessment, Incident Response, Threat Detection, SIEM Operations, Threat Modelling (MITRE ATT&CK, STRIDE, PASTA), Encryption and Key Management, Application and API Security, Role-Based Access Control (RBAC), Audit Logging, Data Integrity (HMAC, Hash Chaining), IT Governance

Networking: Network Design and Installation, Firewall Configuration, VLANs, OSPF, VLSM, SNMP, MQTT, CCNA/CCNP track

Programming and Scripting: Python (FastAPI, pytest), Bash, SQL, HTML/CSS

Platforms and Systems: Linux (Ubuntu, Kali), Windows Server, PostgreSQL, TimescaleDB, EVE-NG, Cisco Packet Tracer

Languages: Arabic (native), English (fluent)

CERTIFICATIONS

Information Security Analyst - Infrastructure and Security

2026

Digital Egyptian Pioneers Initiative (DEPI), MCIT

- Capstone: designed a full **Incident Response Playbook** covering detection, containment, eradication, and recovery phases for enterprise security incidents.
- Core modules: SOC Operations, Vulnerability Management, SIEM and Incident Detection, Cyber Threat Analysis (IoCs and Attack Methodology).

Security Engineer Path - 3 Certificates Completed

2026

TryHackMe

- **Security Engineer:** IAM and cryptography, threat modelling frameworks (MITRE ATT&CK, STRIDE, PASTA), network and system security.
- **DevSecOps:** security in CI/CD, SAST, software composition analysis, secrets and container image scanning, pipeline hardening.
- **Defending AWS:** IAM policy design, VPC network controls, CloudTrail logging, and detection and response for cloud-native attacks.

Google Cybersecurity Professional Certificate

2023

Coursera

- Covered network security, asset and threat management, Python automation for cybersecurity, Linux and SQL for security operations, and detection and response.

VOLUNTEER EXPERIENCE

Junior Teaching Assistant - Network Installation and Maintenance

Sep 2024 – Dec 2024

Zewail City of Science and Technology

Giza, Egypt

- Assisted instructors in lab sessions covering physical network installation, troubleshooting, and maintenance for undergraduate students.

CURRENTLY LEARNING

Preparing for the **AWS Certified Security - Specialty (SCS-C02)** certification (*in progress*).